Bangladesh University of Business and Technology (BUBT)

Project Report

Project Name :Restaurant Billing System

Course Title : Software Development 1

Course Code : CSE 100

Supervised By:

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ABSTRACT

With the arrival of the age of the internet, ***Restaurant Billing*** Sytsem is a ***system*** that computes the total sales of the ***restaurant***, the profit they earn each day and the negative and positive computation of sales versus the target sales.The ***system*** would focus mainly on the development of the ***restaurant billing system*** for Mcdonald's Lipa Ayala itself. It would include the ***systems*** database wherein the computation of sales would be kept in a way that only authorized personnel can access. The ***system*** can also produce report of sales whenever it

DEDICATION

“If you had started doing anything two weeks ago, by today you would have been two weeks better at it.”

— **John Mayer**

Dedicated to our parents, teachers, friends, relatives and all who loved us for all their love and inspirations.

ACKNOWLEDGEMENT

We like to say our gratitude to our creator ALLAH to let us into the world and our parents, who supported us in this whole study and always prayed for our success and good health. We express our deep sense of gratitude to our project Instructor Md. Ashraful Islam for his expert guidance stimulating discussions as well as continued impetus throughout the period of this project and endless patience towards the completion of this project. We feel very proud to work with him. Without the inspiring enthusiasm and encouragement of our supervisor, this work could not have been completed. We thank all the staffs and graduate students at Bangladesh University of Business and Technology (BUBT) and all the friends for their support and encouragement. We would also like to extend our elder and younger brothers. Finally, we wish to express our gratitude to Bangladesh University of Business and Technology (BUBT) for providing an excellent environment for research and all the other facilities to complete the project successfully.

With best regards,

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DECLARATION

I hereby declare that this Project Report titled “Invoice Generation System” submittedto the Department of Computer Science and Engineering, Bangladesh University ofBusiness & Technology, Mirpur-2, Dhaka is a record of original work done by me under the guidance of my supervisor Md. Ashraful Islam Course Instructor, Department of CSE, Bangladesh University of Business & Technology, Mirpur-2, Dhaka. This Project Report is not submitted to any other university or institution for the award of any degree, diplomaor published any time before. I also declare that there has no copy of the source code in the project.

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APPROVAL

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Department of Computer Science and Engineering, Bangladesh University of Business and Technology under the supervision Md. Ashraful Islam, Course Instructor, Department of Computer Science and Engineering has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of Bachelor of Science (B.Sc. Engg.) in Computer Science and Engineering.

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Chapter- 1: Introduction

1.1 Project Aim & Objective

Customer Billing System Project is a basic console application created to show the actual use of the C programming language and its capabilities as well as to develop an application that can be used to charge customers in any department store, shop, café, etc.

You may use this program to keep track of your frequent customers' information such as name, address, phone number, paid amount, due amount, payment date, and so on. You may also add and update accounts at any moment if you have a new client.

The Customer Billing System Project in C source code was created entirely in the C programming language, with no graphics.

1.2 Operation Environment

Programming Language:

- C Programming Language

Compiler:

- GCC (MingW / GNU GCC)

Debugger:

- Interfaces GNU GDB

Chapter- 2: System Analysis

2.1 System Requirement & Specification

■ Ram:

Minimum: 512 MB

Recommended: 1 GB to above

■ Windows:

Minimum: Windows XP

Recommended: Windows 7 to above

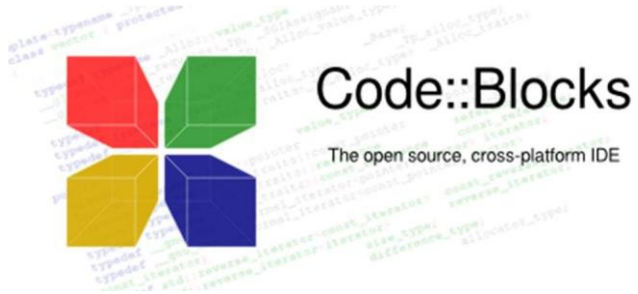
■ Processor:

Minimum: 1 GHz

Recommended: 2 GHz or more

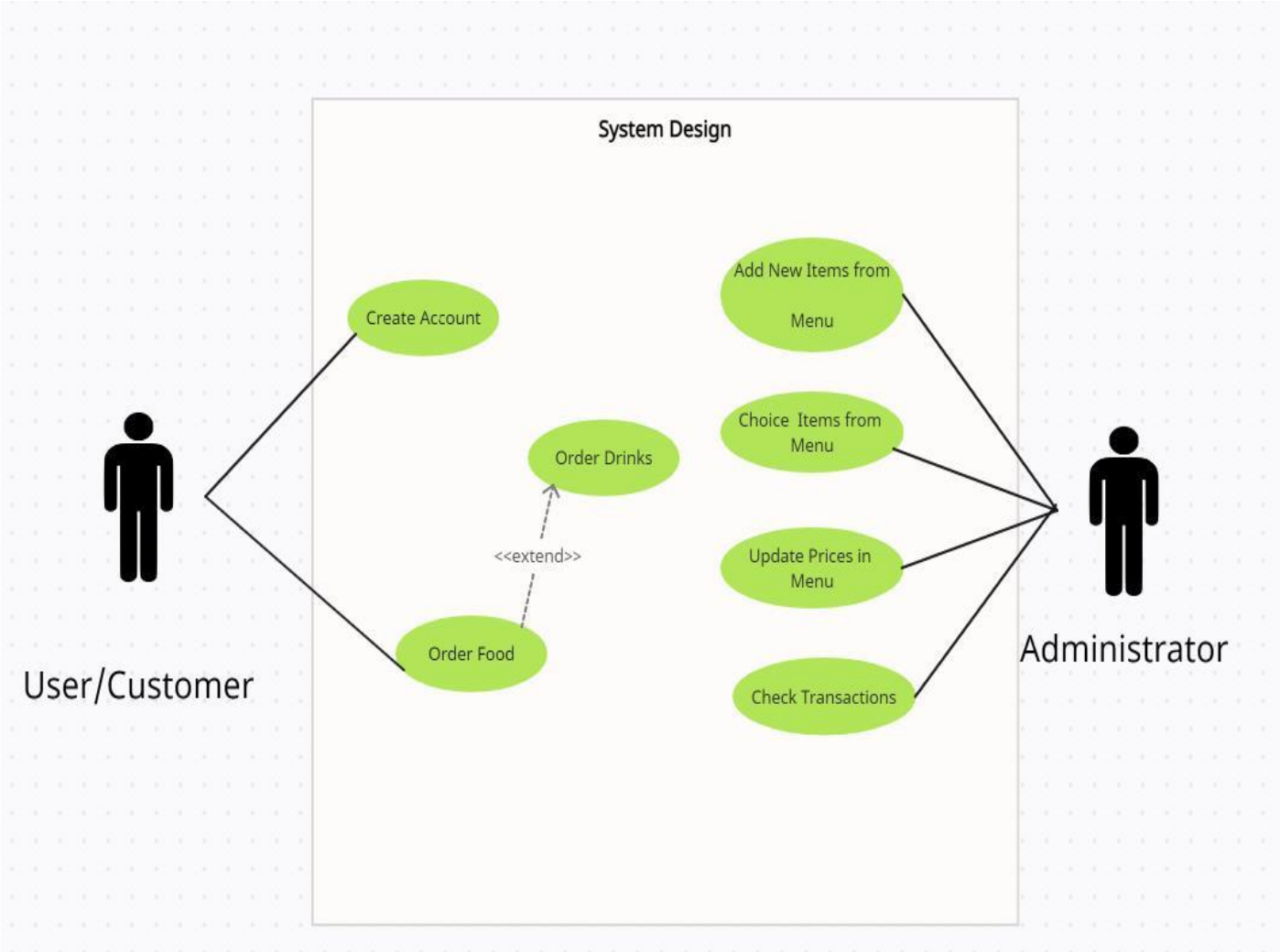
2.2 Software Tools Used

Code::Blocks:



Code::Blocks is a free, open-source cross-platform IDE that supports multiple compilers including GCC, Clang and Visual C++. It is developed in C++ using widgets as the GUI toolkit. Using a plugin architecture, its capabilities and features are defined by the provided plugins. Currently, Code::Blocks is oriented towards C, C++, and Fortran. It has a custom build system and optional Make support. Code::Blocks is being developed for Windows and Linux and has been ported to FreeBSD, OpenBSD and Solaris.

Chapter- 3: System Design



Chapter-4: Coding Analysis

```
#include<stdio.h>
#include<string.h>
#include<stdlib.h>

struct items{
   char item[20];
   float price;
   int qty;
};
struct orders{
   char customer[50];
   char date[50];
   int numOfItems;
struct items itm[50];
};
   void BillHeader(
      char name[50],char date[30]
   )
   {
      printf("\n\n");
      printf("\t    Friend Club Restaurant");
      printf("\n\t   -----------------");
      printf("\nDate:%s",date);
      printf("\nInvoice To: %s",name);
      printf("\n");
      printf("---------------------------------------\n");
      printf("Items\t\t");
      printf("Qty\t\t");
      printf("Total\t\t");
      printf("\n----------------------------------------");
      printf("\n\n");
   }
      void BillBody
      (
         char item[30],
```

```
 int qty,
 float price
 )
 {
 printf("%s\t\t",item);
 printf("%d\t\t",qty);
 printf("%.2f\t\t",qty * price);
 printf("\n");
}
 void BillFooter(float total){
printf("\n");
float netTotal=total;
printf("--------------------------------------\n");
printf("Sub Total\t\t\t%.2f",total);
}
int main(){
    int opt,n;
    struct orders ord;
    struct orders order;
    char saveBill = 'y', contFlag = 'y';
    char name[50];
    FILE *fp;
    while(contFlag == 'y'){
    system("clear");
    float total = 0;
    int invoiceFound = 0;
    printf("\t============Friends Cafe RESTAURANT============");
    printf("\n\nPlease select your prefered operation");
    printf("\n\n1.Generate Invoice");
    printf("\n2.Show all Invoices");
    printf("\n3.Search Invoice");
    printf("\n4.Exit");

    printf("\n\nYour choice:\t");
    scanf("%d",&opt);
    fgetc(stdin);

    switch(opt){
    case 1:
    system("clear");
```

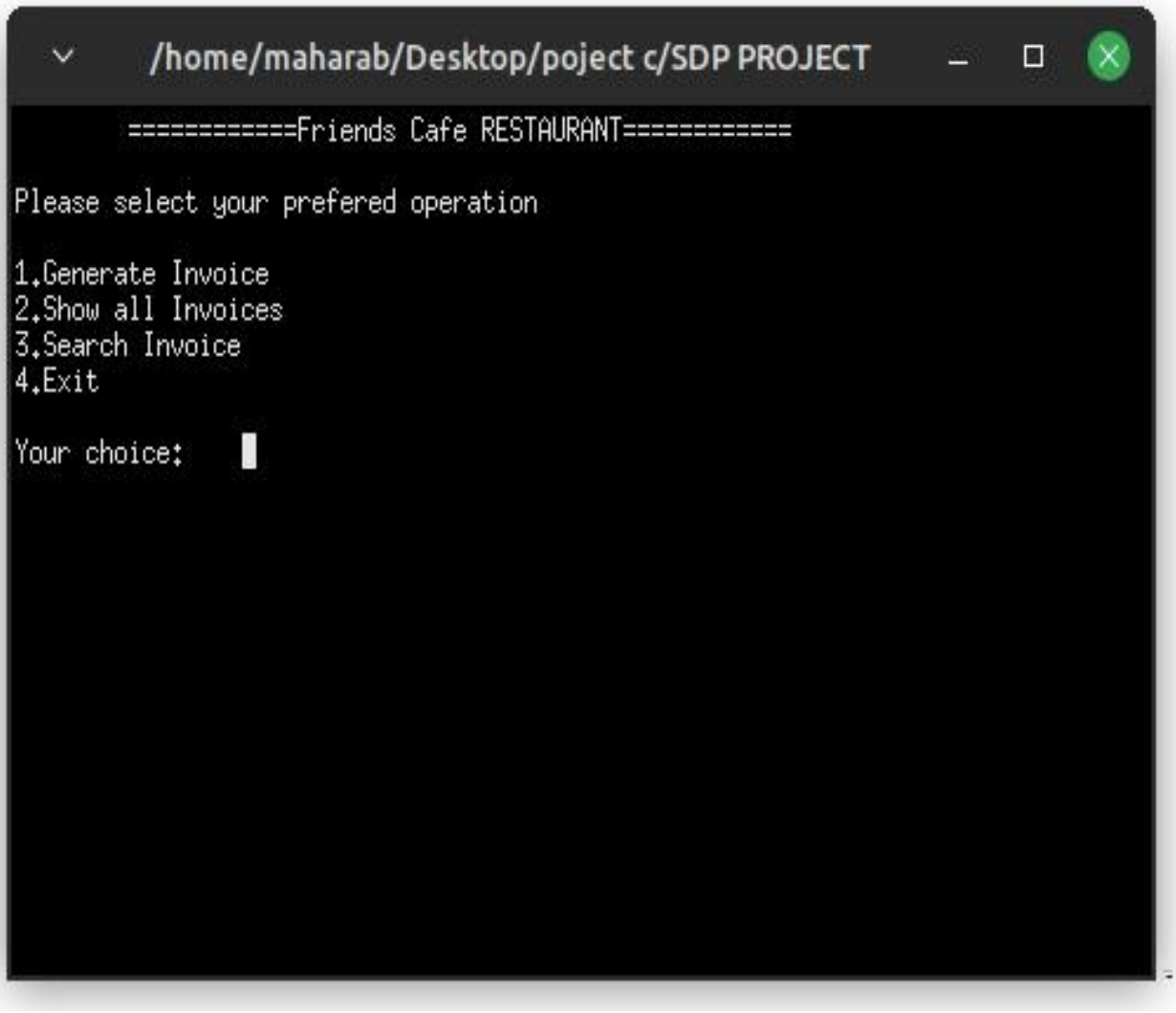
```
  printf("\nPlease enter the name of the customer:\t");
  fgets(ord.customer,50,stdin);
  ord.customer[strlen(ord.customer)-1] = 0;
  strcpy(ord.date,__DATE__);
  printf("\nPlease enter the number of items:\t");
  scanf("%d",&n);
  ord.numOfItems = n;
 for(int i=0;i<n;i++)
{

  fgetc(stdin);
  printf("\n\n");
  printf("Please enter the item Name: %d:\t",i+1);
  fgets(ord.itm[i].item,20,stdin);
  ord.itm[i].item[strlen(ord.itm[i].item)-1]=0;
  printf("Please enter the quantity:\t");
  scanf("%d",&ord.itm[i].qty);
  printf("Please enter the unit price:\t");
  scanf("%f",&ord.itm[i].price);
  total += ord.itm[i].qty * ord.itm[i].price;
}

  BillHeader(ord.customer,ord.date);
  for(int i=0;i<ord.numOfItems;i++){
  BillBody(ord.itm[i].item,ord.itm[i].qty,ord.itm[i].price);
  }
  BillFooter(total);

  printf("\nDo you want to save the bill invoice? (y/n):\t");
  scanf("%s",&saveBill);

  if(saveBill == 'y')
 {
     fp = fopen("Restaurant.txt","a+");
     fwrite(&ord,sizeof(struct orders),1,fp);
     if(fwrite !=0)
     printf("\nInvoice saved successfully!\n");
     else
     printf("\nError saving file!\n");
     fclose(fp);
```

```
        }
        break;
        case 2:
          system("clear");
          fp = fopen("Restaurant.txt","r");
          printf("\n  *****Your Previous Invoices*****\n");
          while(fread(&order,sizeof(struct orders),1,fp)){
          float tot=0;
          BillHeader(order.customer,order.date);
          for(int i=0;i<order.numOfItems;i++){
          BillBody(order.itm[i].item,order.itm[i].qty,order.itm[i].price);
          tot+=order.itm[i].qty*order.itm[i].price;
        }
        BillFooter(tot);
    }
        fclose(fp);
        break;
        case 3:
            printf("Enter the name of the customer:\t");
            fgets(name,50,stdin);
            name[strlen(name)-1] = 0;
            system("clear");

            fp = fopen("Restaurant.txt","r");
            printf("\t*****Invoice of %s*****",name);
            while(fread(&order,sizeof(struct orders),1,fp)){
            float tot = 0;
            if(!strcmp(order.customer,name)){
            BillHeader(order.customer,order.date);
            for(int i=0;i<order.numOfItems;i++){
            BillBody(order.itm[i].item,order.itm[i].qty,order.itm[i].price);
            tot+=order.itm[i].qty * order.itm[i].price;
    }
    BillFooter(tot);
    invoiceFound = 1;
    }

  }
  if(!invoiceFound){
    printf("Sorry the invoice for %s doesnot exists",name);
```

```
        }
        fclose(fp);
        break;
            case 4:
            printf("\n\t\t Bye Bye :)\n\n");
            exit(0);
            break;
            default:
            printf("Sorry invalid option");
            break;
      }
    printf("\nDo you want to perform another operation? [y/n]:\t");
    scanf("%s",&contFlag);
    }
    return 0;
}
```

4.1 Screenshots

This is the opening page of the program.



```
/home/maharab/Desktop/poject c/SDP PROJECT

        ============Friends Cafe RESTAURANT============

Please select your prefered operation

1.Generate Invoice
2.Show all Invoices
3.Search Invoice
4.Exit

Your choice:    1
```

```
/home/maharab/Desktop/poject c/SDP PROJECT

Please enter the name of the customer:  Akash

Please enter the number of items:       3


Please enter the item Name: 1:  Rice
Please enter the quantity:      2
Please enter the unit price:    30


Please enter the item Name: 2:  Chicken Curry
Please enter the quantity:      2
Please enter the unit price:    120


Please enter the item Name: 3:  Fish Curry
Please enter the quantity:      1
Please enter the unit price:    120


            Friend Club Restaurant
           -----------------
Date:May 27 2023
Invoice To: Akash
-------------------------------------
Items           Qty             Total
-------------------------------------

Rice            2               60.00
Chicken Curry           2               240.00
Fish Curry              1               120.00

-------------------------------------
Sub Total                       420.00
Do you want to save the bill invoice? (y/n):    y

Invoice saved successfully!
```

```
/home/maharab/Desktop/poject c/SDP PROJECT

      ============Friends Cafe RESTAURANT============

Please select your prefered operation

1.Generate Invoice
2.Show all Invoices
3.Search Invoice
4.Exit

Your choice:    2
```

```
                Friend Club Restaurant
               -----------------
Date:May 27 2023
Invoice To: Akash
-------------------------------------
Items            Qty              Total
-------------------------------------

Rice             2                60.00
Chicken Curry            2                240.00
Fish Curry               1                120.00

-------------------------------------
Sub Total                         420.00
Do you want to perform another operation? [y/n]:
```

```
Do you want to perform another operation? [y/n]:       y
```

.

```
        ============Friends Cafe RESTAURANT============

Please select your prefered operation

1.Generate Invoice
2.Show all Invoices
3.Search Invoice
4.Exit

Your choice:    4

                 Bye Bye :)


Process returned 0 (0x0)   execution time : 822.883 s
Press ENTER to continue.
```

Chapter- 5: Conclusion

5.1 Limitations

For our work we used updated software and technologies. Still our proposed system mayface some drawbacks and some are listed below:

- Our system can’t verify user information.

- Our system is for bill payment, it can be done by user. But we need another software for online payment system.

5.2 Future works

We have many user friendly features and modification for the project in the near future. Which we couldn't done this time due to lack of manpower & fund. We can achieve moreaccuracy by fixing the limitations of our program. Some works which we can improve are given below:

- We will add a pay bill feature. In this feature user can pay bill to various companyand services.
- We will add a mobile recharge feature that will help user easily recharge balance intheir phone number by few clicks.
-
-

5.3 Conclusion

A billing system includes procedures and processes that help create bills and invoices for customersThe statement provides important information such as the total amount due, due date, minimum payment required, and any applicable fees or charges. Customers can use the billing statement to track their expenses, monitor their payments, and reconcile their accounts.

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REFRENCES

Here are some website links which we used for our project.

- **https://github.com/**
- **https://www.codewithc.com/**
- **https://code-projects.org**
- **https://www.wikipedia.org/**
- **https://www.google.com/**
- **Shttps://www.w3schools.com/**